Multi-channel Source Separation 1-2

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Speech And Audio Recognition
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https://mairlab-km.github.io/

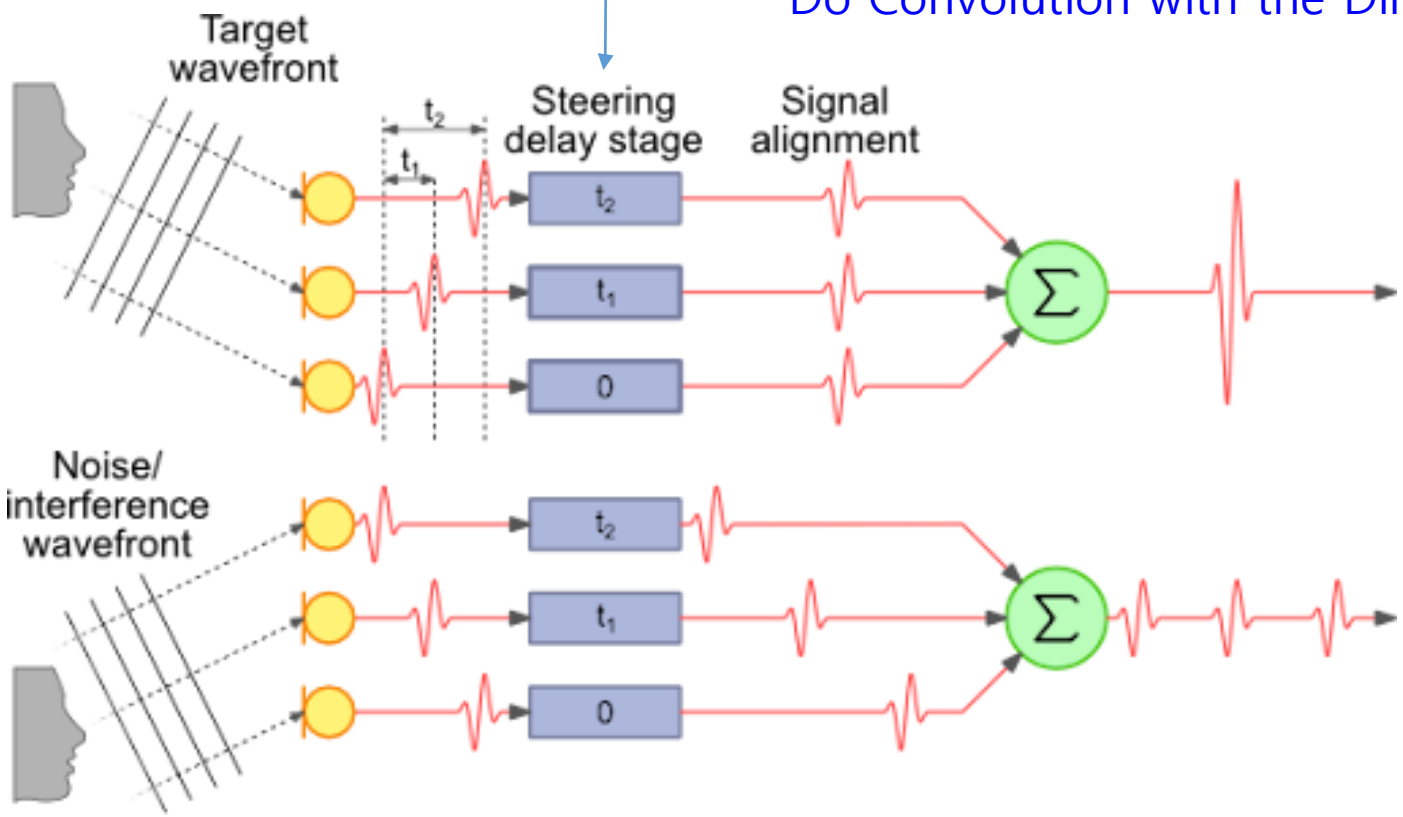
This lecture material refers to
https://github.com/yandexdataschool/speech_course?tab=readme-ov-file and
https://github.com/markovka17/dla

Multi-channel Source Separation

- Delay-and-Sum Beamformer
  - Consider the propagation delays!

$$g(t) = \delta(t + t_1)$$

Do Convolution with the Dirac delta function!



https://medium.com/@itberrios6/introduction-to-beamforming-part-1-92767408f483

Multi-channel Source Separation

- Steering vector: a time delay becomes a phase shift

Delay

$$\tau_m(\theta) = \frac{(m-1)\, d \sin\theta}{c}$$

extra travel time to mic m (plane wave, mic spacing d)

Steering vector

$$\boldsymbol{h}(\theta, \omega) = [e^{-j\omega\tau_1(\theta)}, \ \ldots, \ e^{-j\omega\tau_M(\theta)}]^T$$

what the array receives from direction θ: x(ω) = h(θ, ω)·S(ω)

DAS

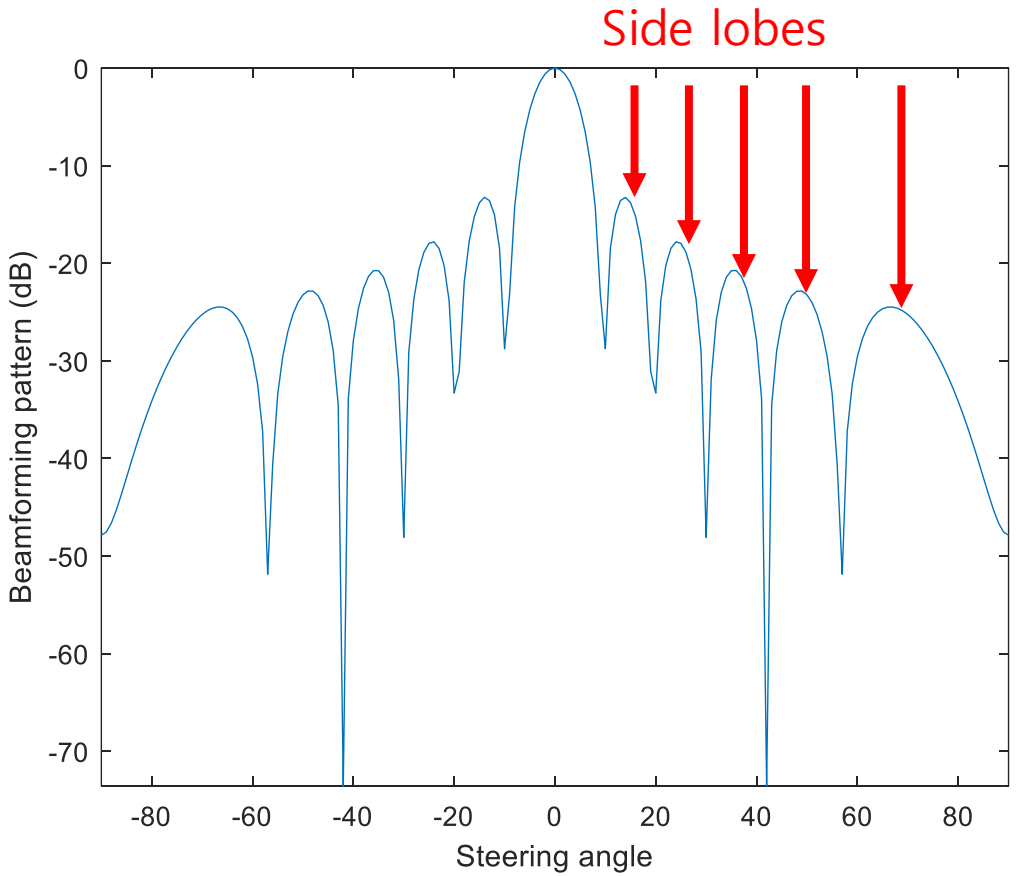
$$\boldsymbol{w}_{\mathrm{DAS}} = \frac{\boldsymbol{h}}{M}, \qquad y = \boldsymbol{w}^H \boldsymbol{x} = \frac{1}{M} \sum_{m=1}^{M} e^{+j\omega\tau_m} X_m(\omega)$$

why h / M ? → next slide

Steered to the source (θ = θ₀): $\mathbf{w}^H\mathbf{h} = 1$ → y(ω) = S(ω)

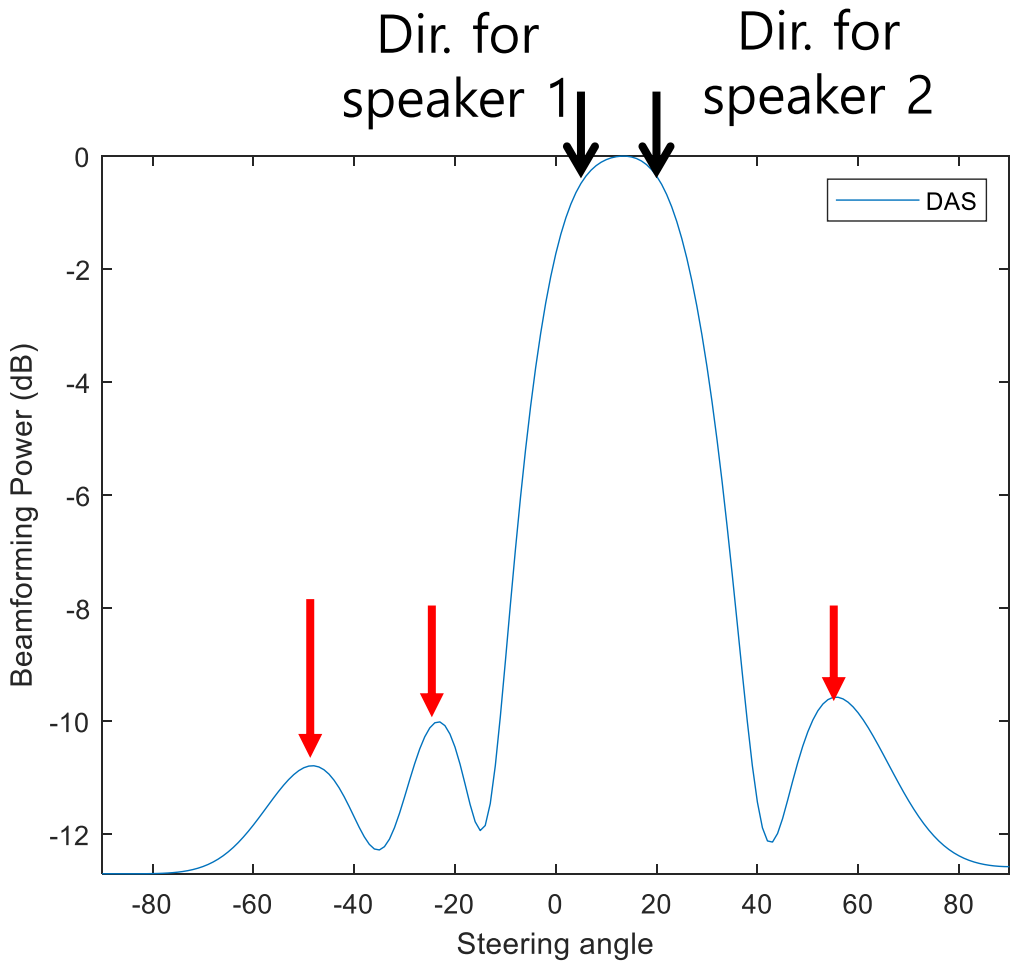
Multi-channel Source Separation

- Two problems of the Delay-and-Sum Beamformer
  - ① Side lobes: other directions leak in (≈ -13 dB)

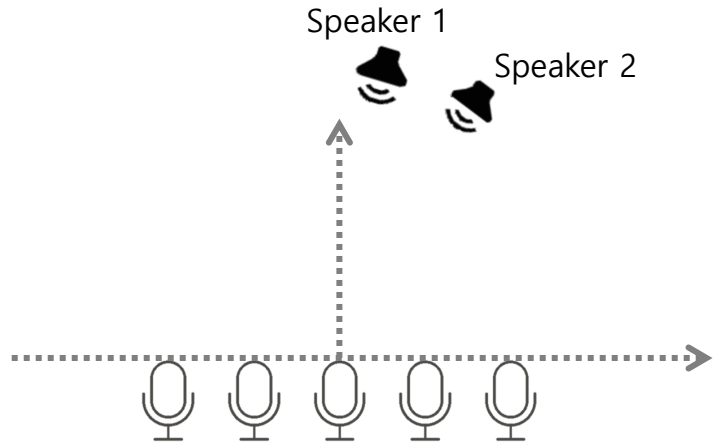


Multi-channel Source Separation

- Two problems of the Delay-and-Sum Beamformer
  - ② Main lobe too wide: both speakers fall inside it

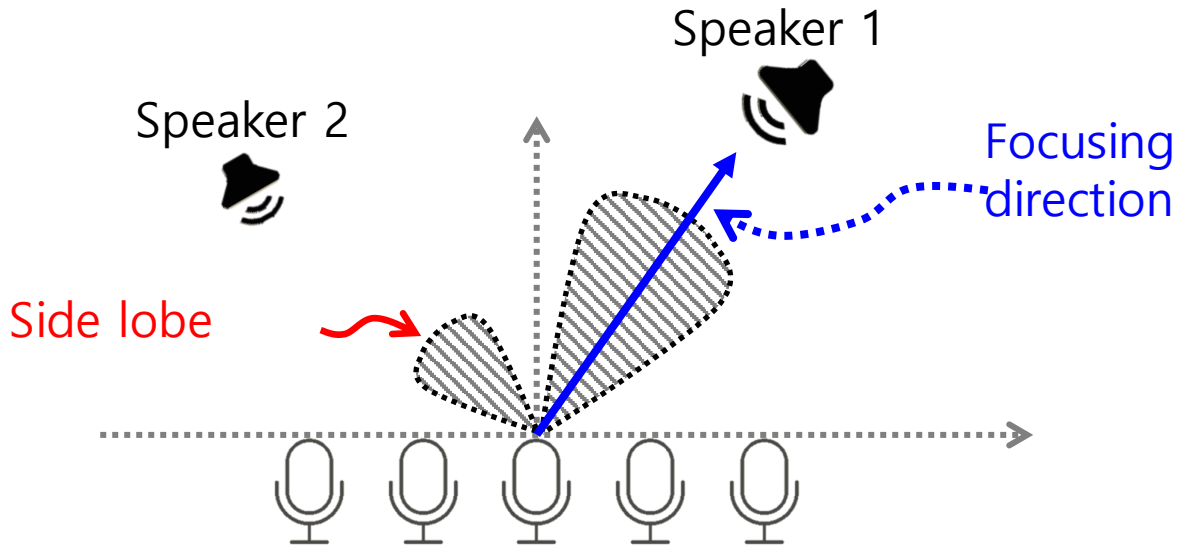


Delay-and-Sum beamformer



Multi-channel Source Separation

- The problem of the Delay-and-Sum Beamformer
  - DAS weights are fixed: they cannot adapt to the interference

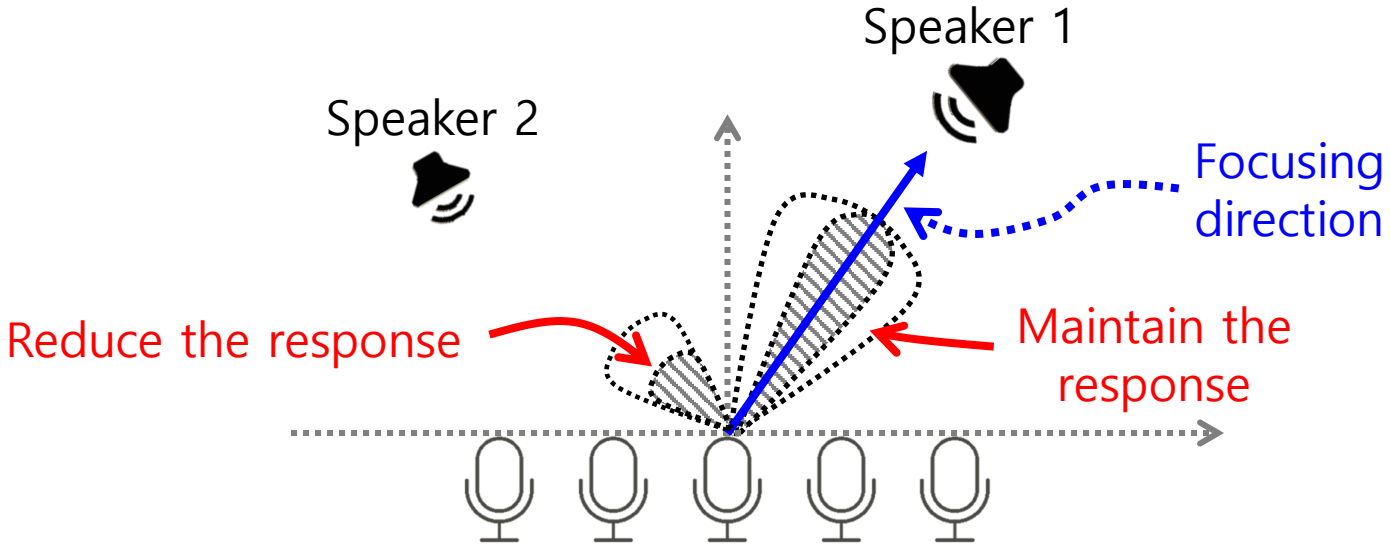


  - How to reduce the response for the speaker 2? when focusing on the speaker 1...

→ **MVDR beamformer!**

Multi-channel Source Separation

- MVDR beamformer
  - Design a spatial filter w that only responds to a specific direction



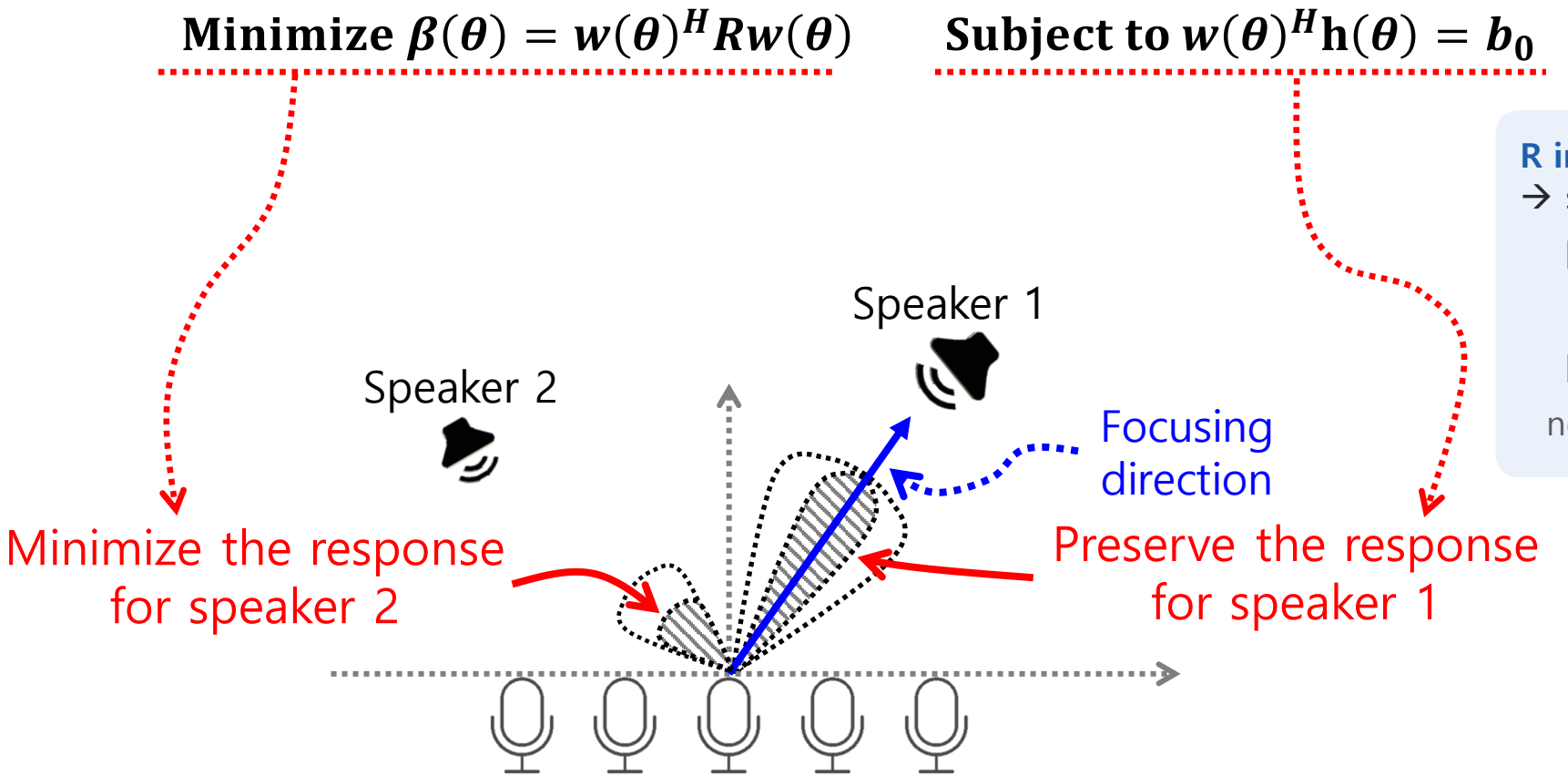
  - Design a spatial filter w that minimizes the total beamforming power but maintain the response in the aiming angle

$$\textbf{Minimize } \boldsymbol{\beta} = \boldsymbol{w}^{\boldsymbol{H}} E[\boldsymbol{x}(\boldsymbol{\omega})\boldsymbol{x}(\boldsymbol{\omega})^{\boldsymbol{H}}]\boldsymbol{w} \qquad \textbf{Subject to } \boldsymbol{w}^{\boldsymbol{H}}\mathbf{h} = \boldsymbol{b_0}$$

Plane wave model

Multi-channel Source Separation

- MVDR beamformer
  - Design a spatial filter w that minimizes the total beamforming power but maintain the response in the aiming angle



R includes the target
→ strictly, MPDR

MPDR $R_x = E[xx^H]$
includes the target

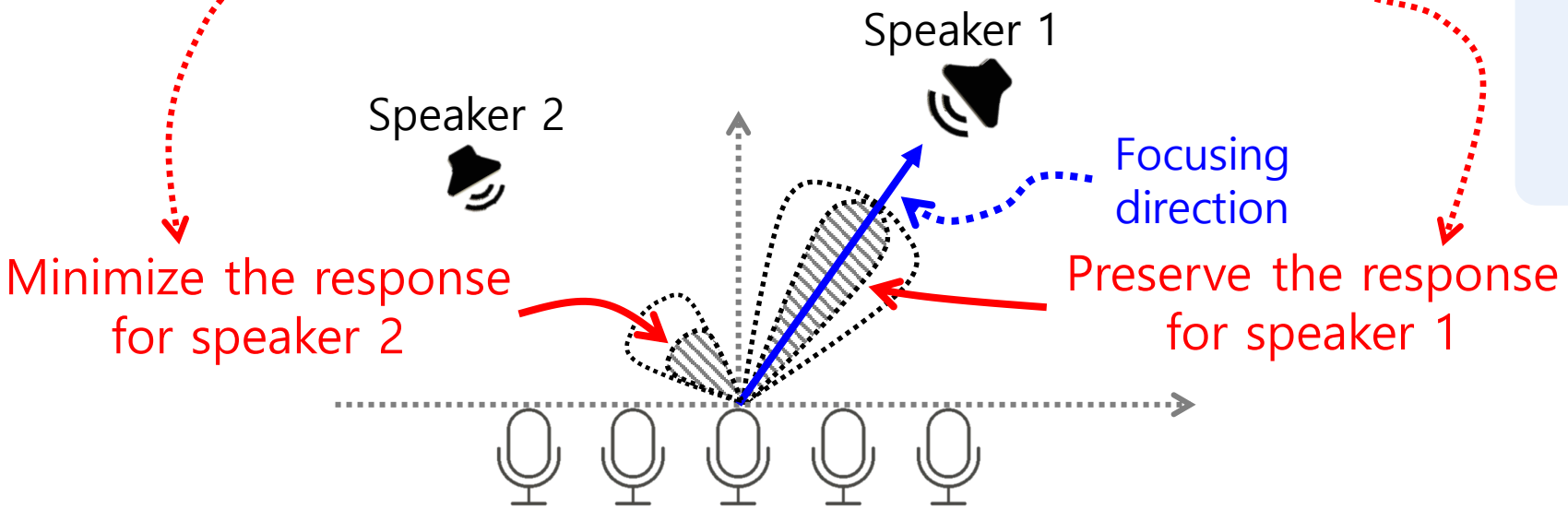
MVDR $R_n = E[nn^H]$
noise + interference only

Multi-channel Source Separation

- MVDR beamformer
  - Design a spatial filter w that minimizes the total beamforming power but maintain the response in the aiming angle

$$\textbf{Minimize } \boldsymbol{\beta}(\boldsymbol{\theta}) = \boldsymbol{w}(\boldsymbol{\theta})^{\boldsymbol{H}}\boldsymbol{R}\boldsymbol{w}(\boldsymbol{\theta}) \qquad \textbf{Subject to } \boldsymbol{w}(\boldsymbol{\theta})^{\boldsymbol{H}}\mathbf{h}(\boldsymbol{\theta}) = \boldsymbol{b_0}$$

$$\textit{solve} \Rightarrow \boldsymbol{w}(\boldsymbol{\theta}) = \frac{\boldsymbol{R}^{-1}\mathbf{h}(\boldsymbol{\theta})}{\mathbf{h}(\boldsymbol{\theta})^{\boldsymbol{H}}\boldsymbol{R}^{-1}\mathbf{h}(\boldsymbol{\theta})}\boldsymbol{b}_0^*$$



R includes the target
→ strictly, MPDR

MPDR $\boldsymbol{R}_x = E[\boldsymbol{x}\boldsymbol{x}^H]$
includes the target

MVDR $\boldsymbol{R}_n = E[\boldsymbol{n}\boldsymbol{n}^H]$
noise + interference only

Multi-channel Source Separation

- MVDR solution via a Lagrange multiplier (참고)

① **Lagrangian**

$$\mathcal{L} = \boldsymbol{w}^H \boldsymbol{R} \boldsymbol{w} + \lambda^* (\boldsymbol{w}^H \boldsymbol{h} - 1) + \lambda (\boldsymbol{h}^H \boldsymbol{w} - 1)$$

② **∂L/∂w* = 0**

$$\boldsymbol{R}\boldsymbol{w} + \lambda^* \boldsymbol{h} = \boldsymbol{0} \;\Rightarrow\; \boldsymbol{w} = -\lambda^* \boldsymbol{R}^{-1} \boldsymbol{h}$$

③ **$w^H h = 1$**

$$\lambda = -\frac{1}{\boldsymbol{h}^H \boldsymbol{R}^{-1} \boldsymbol{h}}$$

④ **Solution**

$$\boldsymbol{w}_{\mathrm{MVDR}} = \frac{\boldsymbol{R}^{-1} \boldsymbol{h}}{\boldsymbol{h}^H \boldsymbol{R}^{-1} \boldsymbol{h}}$$

Check: $R = \sigma^2 I$ → $w_{MVDR} = h / M = w_{DAS}$

Multi-channel Source Separation

- What does $w^H Rw$ measure?

Output power of the beamformer

$$\beta = E[|y|^2] = E[|\boldsymbol{w}^H \boldsymbol{x}|^2] = \boldsymbol{w}^H E[\boldsymbol{x}\boldsymbol{x}^H]\,\boldsymbol{w} = \boldsymbol{w}^H \boldsymbol{R}\,\boldsymbol{w}$$

Correlation matrix R (M × M, one per frequency bin)

$$\boldsymbol{R} = E[\boldsymbol{x}\boldsymbol{x}^H] \approx \frac{1}{T}\sum_{t=1}^{T} \boldsymbol{x}\,(t,f)\,\boldsymbol{x}(t,f)^H$$

estimated by averaging over the STFT frames of the recording

Uncorrelated sources: each source adds its own term

$$\boldsymbol{R} = \sigma_1^2\,\boldsymbol{h}_1\boldsymbol{h}_1^H + \sigma_2^2\,\boldsymbol{h}_2\boldsymbol{h}_2^H + \sigma_n^2\,\boldsymbol{I}$$

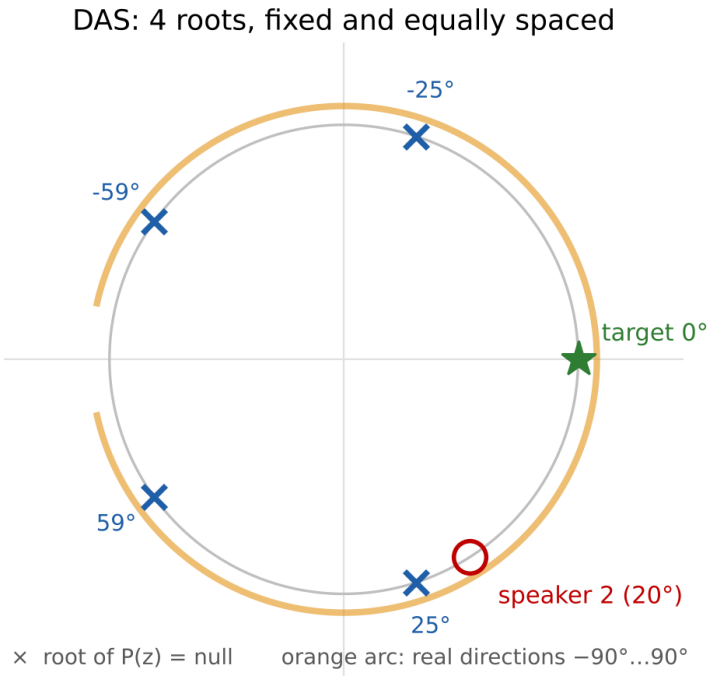
speaker 1 + speaker 2 + sensor noise

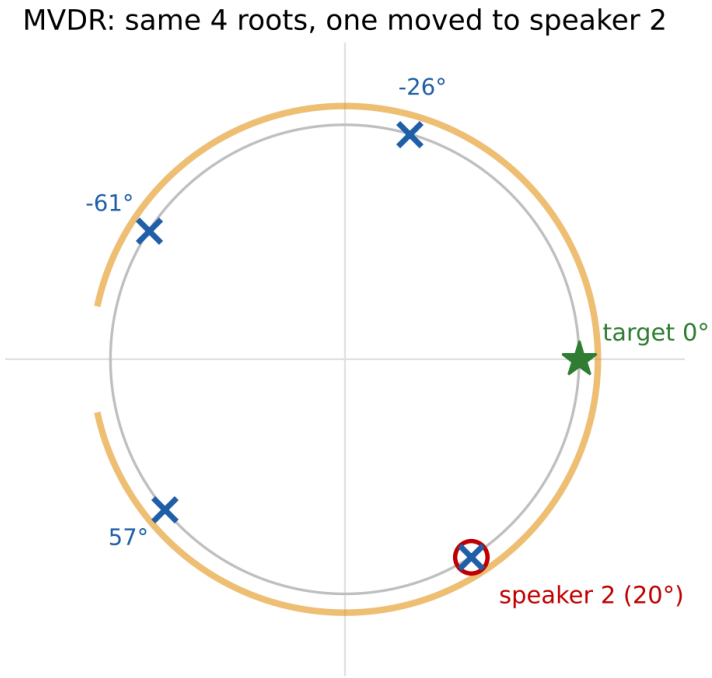
Multi-channel Source Separation

- How many nulls can one filter make?

$$A(\theta) = \boldsymbol{w}^H \boldsymbol{h}(\theta) = \sum_{m=0}^{M-1} w_m^* \, z^m = P(z), \qquad z = e^{-j\phi}, \;\; \phi = \frac{\omega d \sin\theta}{c}$$

Polynomial of degree M−1 → at most M−1 roots(제곱근) → **at most M−1 nulls**





Multi-channel Source Separation

- Counting conditions: K sources need K ≤ M

$$\underbrace{\boldsymbol{w}^H \boldsymbol{h}_1 = 1}_{\text{keep target: } 1} \qquad \underbrace{\boldsymbol{w}^H \boldsymbol{h}_k = 0,\ k = 2, \ldots, K}_{\text{nulls: } K-1} \qquad \underbrace{\boldsymbol{w} \in \mathbb{C}^M}_{\text{unknowns: } M} \quad \Rightarrow \quad \boldsymbol{K - 1 \leq M - 1}$$

Solvable if the K steering vectors are **linearly independent** — not necessarily orthogonal ($\mathrm{h}^H \cdot \mathrm{h} = 0$)

$$\rho = \frac{\left|\boldsymbol{h}_2^H \boldsymbol{h}_1\right|}{M} \quad \text{(similarity of target and interferer)}$$

ρ = 0 : orthogonal
- null for free
- ||w||² = DAS

0 < ρ < 1 : independent
- null possible
- ||w||² grows × 1/(1−ρ²)

ρ = 1 : same vector
- same direction or grating lobe
- null impossible

Multi-channel Source Separation

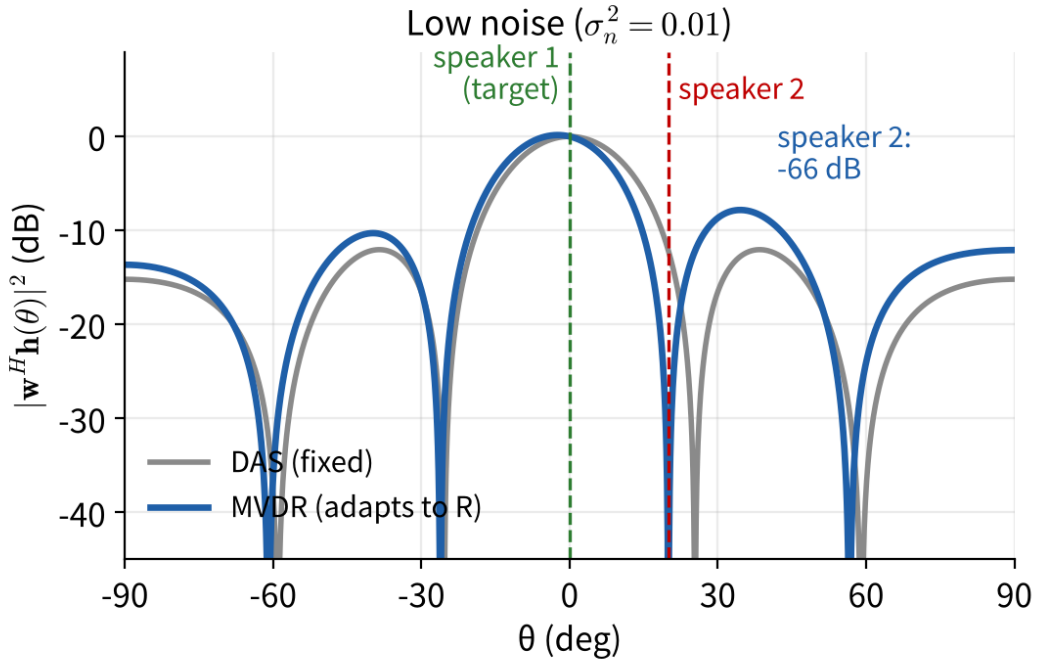
- Why does minimizing $w^H Rw$ create a null?

$$\beta = \boldsymbol{w}^H \boldsymbol{R} \boldsymbol{w} = \sigma_1^2 |\boldsymbol{w}^H \boldsymbol{h}_1|^2 + \sigma_2^2 |\boldsymbol{w}^H \boldsymbol{h}_2|^2 + \sigma_n^2 \| \boldsymbol{w} \|^2$$

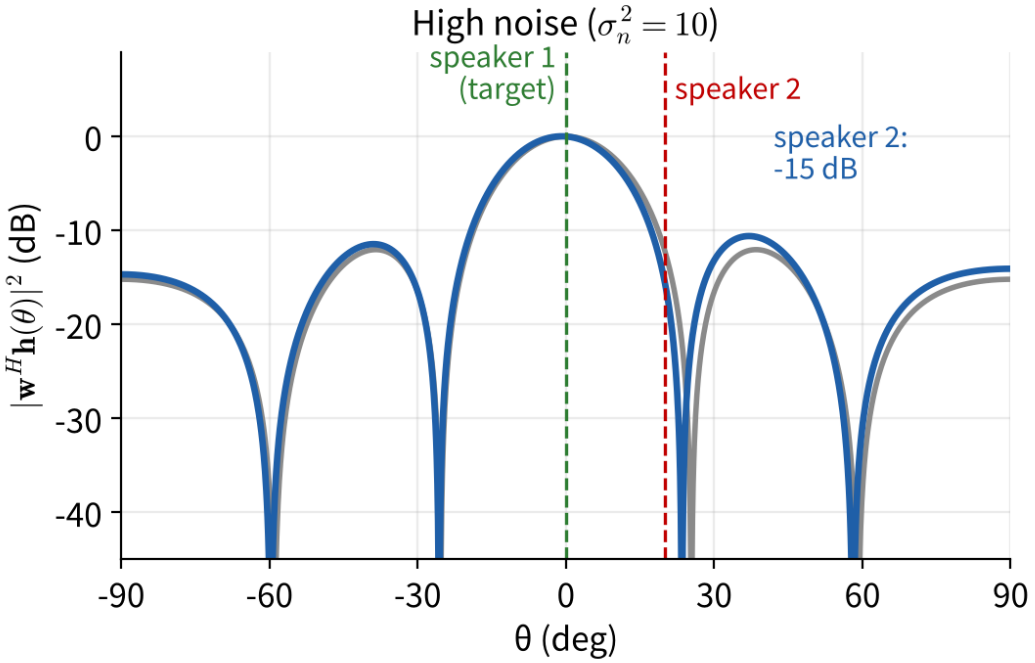
σ_1^2 fixed by $\boldsymbol{w}^H \boldsymbol{h}_1 = 1$

0 : null toward speaker 2

noise



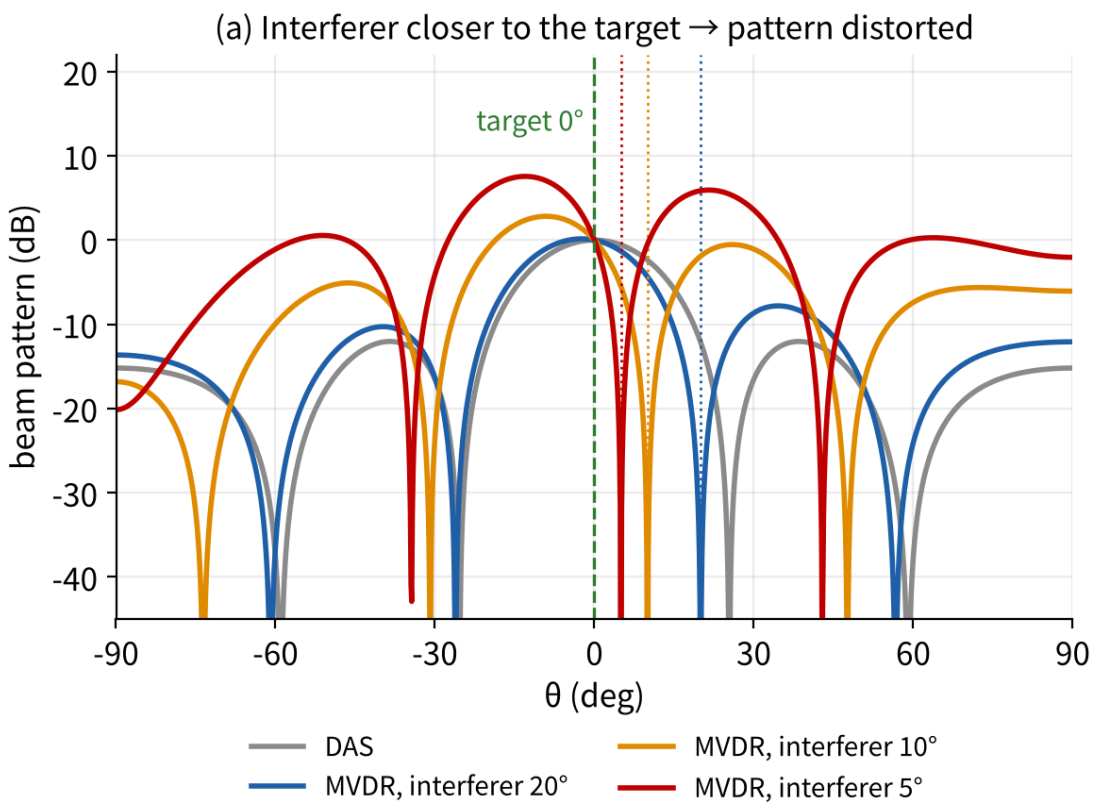
<Beam pattern>

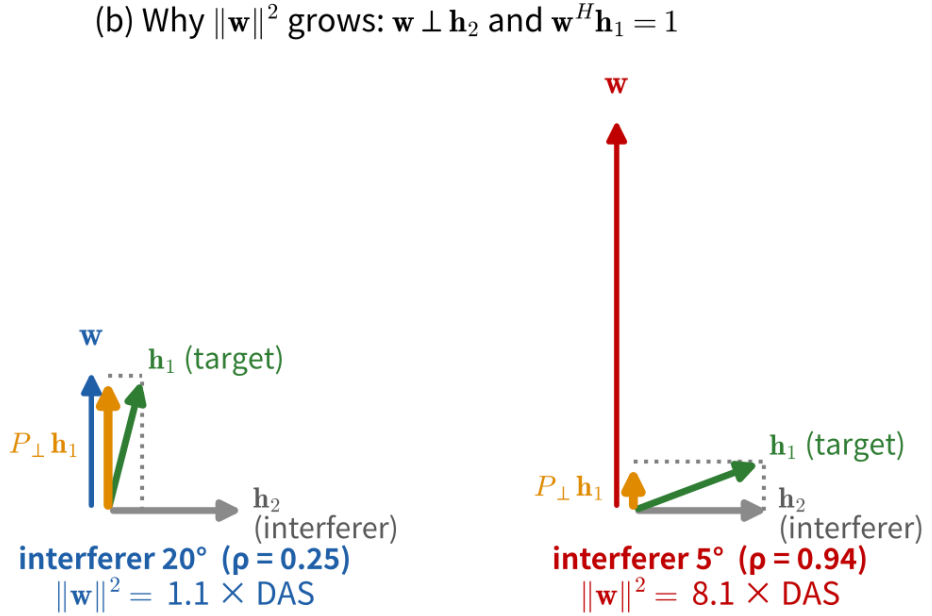


<Beam pattern>

Multi-channel Source Separation

- Deeper nulls are not free

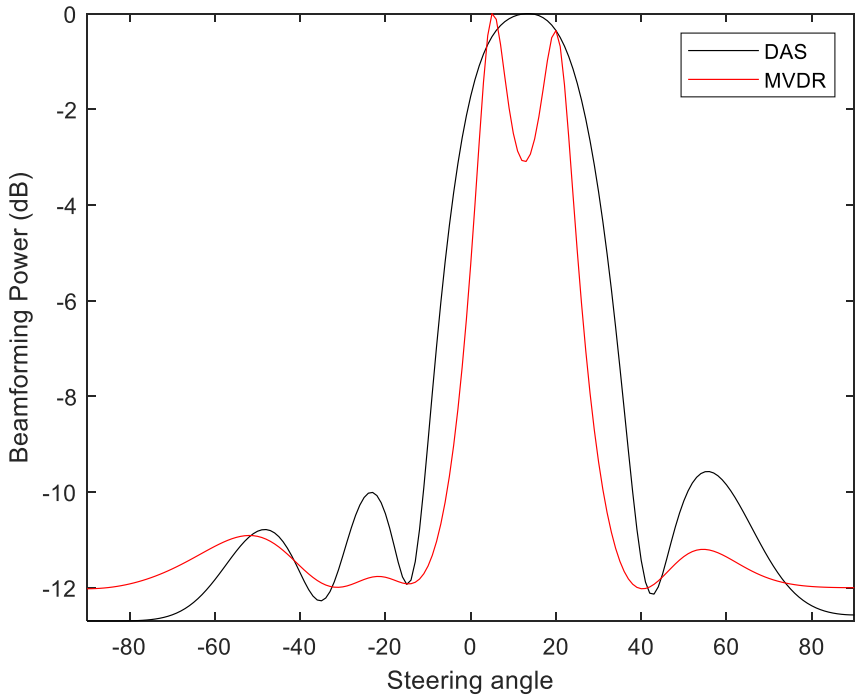




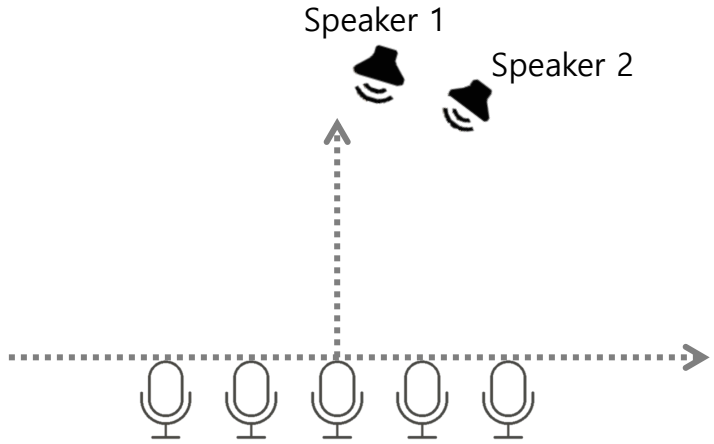
1. Noise amplified
2. Pattern distorted
3. Narrow null misses on small errors

Multi-channel Source Separation

- MVDR beamformer
  - Scanning θ → spatial power spectrum = DoA estimator







$$P_{\mathrm{DAS}}(\theta) = \frac{\boldsymbol{h}(\theta)^H \boldsymbol{R}\, \boldsymbol{h}(\theta)}{M^2}$$

$$P_{\mathrm{MVDR}}(\theta) = \frac{1}{\boldsymbol{h}(\theta)^H \boldsymbol{R}^{-1} \boldsymbol{h}(\theta)}$$

MVDR: side lobes suppressed, two speakers resolved
(Beamforming Power: 조향각(θ) 마다 필터($w(\theta)$)를 만들고, 그 때의 출력 파워($P(\theta)$)를 그린 것)

Multi-channel Source Separation

- The limitation of the beamformer!
  - We have to know the direction-of-arrival (DoA) of the sound

$$y(\theta,t)=\sum_{m=1}^{M} g_m(\theta,t) * x_m(t) \quad \Longleftrightarrow \quad y(\theta,\omega)=\boldsymbol{w}(\theta,\omega)^H \boldsymbol{x}(\omega)$$

Convolution

FFT & iFFT

Beamforming output | Weight (Spatial filter) | Measured pressure signals

  - However, an accurate DoA is hard to obtain (noise, reverberation, multiple sources)
  - **→ Lec 9: separation without DoA (BSS, DNN-based beamforming)**